Detecting Fake News and Misinformation on Social Media Platform

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1 Problem Idea

Social media is heavily integrated into our daily lives. It is so well integrated that it has become a primary source of information for users to know about the world, the news, their friends, advice, hacks, trends, and much more. This is due to the ease of accessibility and the way social media platforms allow users to stay up to date in real time. With how quickly information can spread across the world, this poses questions and concerns on whether the information that we see on our devices is credible or accurate. The problem we are investigating is detecting misleading or false information, commonly known as fake news, through social media platforms, specifically on Twitter. We believe that this problem is important as it can negatively impact the perceptions of the people, cause people to become weary and lose their trust, and bury the truth. As misinformation can be developed and spread as fast as legitimate information, we believe it is critical to propose or develop methods that could identify misinformation within social media platforms and potentially mitigate it. We believe that doing so with this topic could ensure credibility and reliability within posts and content, along with helping users gain or regain that trust in getting their news from social media platforms. This topic is related to social media analysis, as it analyzes patterns within the message or content of the post and how users interact with that message or content. Specifically, we are planning to look at linguistic patterns within the messages or post, potentially finding the source of the misinformation, and examining different strategies and techniques to identify and filter misinformation.

2 Motivation

The rapid spread of misinformation online has become an increasing worry as social media usage rises. Platforms such as Facebook and Twitter are designed to support immediate interaction and effortless content sharing across large networks. While this design increases user engagement, it also enables misinformation to spread widely before it can be verified.

Highlighting the extent of this issue is a study conducted by researchers at Penn State University, where they analyzed millions of posts containing links shared on Facebook between 2017 and 2020. The study found that approximately 75% of shares were made without users clicking the link first [1]. This behavior suggests that users tend to share information without fact-checking or evaluating its credibility.

This issue is especially concerning given that social media has become a primary source of news and information for many individuals, and the consequences of misinformation spread into various domains. In political contexts, misleading content has been shown to influence public perception during elections and major events [2]. In healthcare, misinformation about vaccines and disease treatments has affected personal health decisions and trust in science [3].

On social media platforms, misinformation appears in short-form posts rather than full-length articles. These posts often rely on concise and striking language in order to increase their chances at being liked and shared. As a result, typical approaches that focus on analyzing full articles or long-form content may not be sufficient for detecting misinformation. This creates a need for methods that can operate directly at the level of social media posts. This project is motivated by the goal of developing a model capable of detecting fake news and misinformation within social media content, providing a more appropriate approach for the way information is created and shared on these platforms.

3 Dataset

The dataset that we plan to use is the CIC truth seeker dataset. This dataset is from the Canadian Institute for Cybersecurity, and it aims to be the largest ground-truth fake news analysis dataset in relation to social media. It contains more than 180,000 labeled tweets, all of which were verified using a three-factor verification method with 456 Amazon Mechanical Turkers for labeling each tweet.

This dataset is appropriate for our research because it not only provides the actual text, but it also includes a number of lexical features like the number of present verbs, past verbs, and adjectives, as well as text features like the number of unique, complex words. It also introduces bot, credibility, and influence scores to better help understand patterns of Twitter users. All of this combined gives us a wide range of information to create a robust system.

3.1 Dataset Examples

Key variables include:

- **author**: Source of the original statement
- **statement**: News headline or claim being evaluated
- **tweet**: A Twitter post related to the statement
- **BinaryNumTarget**: Ground-truth label (1 = True, 0 = False)

- **manual_keywords**: Keywords used to retrieve related tweets
- **3_label_majority_answer / 5_label_majority_answer**: Crowd-sourced agreement labels

Example data instances:

- **Example 1**
  - **BinaryNumTarget**: 1 (True)
  - **statement**: Trump, federal minimum wage
  - **tweet**: @MichaelTuckness @LeeStls33 @Interloper_X @Nicole-Marie1081 @restore_rescue @DonaldJTrumpJr Biden released his 2021 tax returns after Trump refused to release his for his entire 4-year term.
    Biden increased the minimum wage for thousands of federal contractors to $15/hr.
    Biden directed $2.5 billion to address America's mental illness and addiction crisis.
  - **3_label_majority_answer**: Mostly Agree
  - **5_label_majority_answer**: Agree
- **Example 2**
  - **BinaryNumTarget**: 0 (False)
  - **author**: Tom Kertscher
  - **statement**: Joe Biden's great-grandfather Joseph J. Biden was a slave owner and fought for the Confederacy.
  - **tweet**: So Kamala Harris wants to be Biden's VP, yet she said she believed his accusers—what a hypocrite. She also supports BLM, yet forgets to mention that her great-grandfather was one of the largest slave owners in Jamaica 80 years after the US banned slavery.
  - **manual_keywords**: Biden, great grandfather, slave owner
  - **3_label_majority_answer**: Agree
  - **5_label_majority_answer**: Agree
- **Example 3**
  - **BinaryNumTarget**: 1 (True)
  - **author**: Ciara O'Rourke
  - **statement**: Says Spectrum will provide free internet to students during coronavirus school closures.
  - **tweet**: Spectrum giving two months of free internet for college students—that is $100 saved, I need that.
  - **manual_keywords**: Spectrum, free internet, students
  - **3_label_majority_answer**: No Majority
  - **5_label_majority_answer**: Agree

4 Approach

For this project, we assess various types of text representations as well as various classification methods in order to detect fake news on social media.

First, we perform text processing on the tweets using text cleaning operations such as lower casing, removal of punctuation marks, and tokenization.

Then, we try out various text representation techniques, including TF-IDF, which is commonly used to extract word frequency features from a document. Other methods include encoding semantics through Word2Vec and GloVe, along with pre-trained BERT representation for contextual semantics. We will also be testing LLM prompting with Llama 3.1 8B to see how results hold up to traditional ML methods.

Table 1: Experimental Setup

Exp.	Representation	Model
1	TF-IDF	Logistic Regression
2	Word2Vec	Random Forest
3	GloVe	Decision Tree
4	BERT (fine-tuned)	Classification Head
5	BERT (embeddings)	SVM
6	Raw Text	LLM Prompting

From Table 1, it can be observed that 6 different combinations of representation techniques and classifiers are tested. Simple baseline representations such as Term Frequency-Inverse Document Frequency (TF-IDF), Word2Vec, and GloVe are combined with linear and non-linear classifiers to test their effectiveness. BERT is used for obtaining contextualized representations using fine-tuning and feature extraction. Lastly, an LLM is used to compare and contrast with traditional ML approaches.

In order to obtain reliable results, a train-test split is applied with a fixed random state. The goal here is to determine the optimal combination for detecting misinformation on short-form social media platforms.

5 Metrics

To evaluate our fake news detection models, we use several standard classification metrics. Accuracy measures the overall correctness of predictions, but it may be misleading if the dataset is imbalanced.

Precision measures how many predicted fake news instances are actually fake, and recall measures how many real fake news instances are successfully detected. Since both false positives and false negatives are important in this task, we also use the F1-score to balance precision and recall.

In addition, we use a confusion matrix to analyze prediction errors and ROC-AUC for models that output probabilities to evaluate their ability to distinguish between true and false information.

6 Results

See the final page for Table 2 and Figure 1. Table 2 compares the methodology with accuracy, precision, recall, F1-score, and ROC-AUC. Figure 1 shows the confusion matrices for all six methods. The confusion matrices were done with a sample size of 2000, except for LLM, which used 1000.

6.1 TF-IDF + Logistic Regression

The TF-IDF model achieved strong overall performance, with an accuracy of 0.9575 and a macro F1-score of 0.9574. This shows that the model performed well on both fake and true news classes rather than only favoring one class. For the true class, the model reached a precision of 0.9507, a recall of 0.9681, and an F1-score of 0.9593, meaning it correctly identified most true news instances. The fake class also performed well, with a precision of 0.9651, a recall of 0.9461, and an F1-score of 0.9555.

From the confusion matrix, the model correctly classified 912 fake tweets and 1003 true tweets. It made 52 false positive errors, where fake tweets were incorrectly predicted as true, and 33 false negative errors, where true tweets were incorrectly predicted as fake. Since the number of errors is relatively small compared to the total test size of 2000, this suggests that TF-IDF is effective for detecting fake news in this dataset.

The ROC-AUC score of 0.9873 indicates that the model has excellent ability to distinguish between fake and true news. A value close to 1.0 means the classifier is very effective at separating the two classes.

TF-IDF likely performs strongly because fake and true tweets may use different keyword patterns, phrasing, or repeated terms. Since the project focuses on misinformation in short-form social media posts, word-level features can be especially useful because tweets often rely on concise and attention-grabbing language. This matches the project motivation that misinformation on social media appears in short posts rather than long articles, making text-level patterns important for detection.

6.2 Word2Vec + Random Forest

The Word2Vec model performed well with an accuracy score of 0.8240, a precision score of 0.8314, a recall score of 0.8282, a F1 score of 0.8298, and a ROC-AUC score of 0.8997. These scores it shows that this method was able to identify fake news and authentic news, as supported by the ROC-AUC value being close to 1.0.

As seen in the confusion matrix, the ratio between the amount of correct predictions for both fake and authentic news is pretty even, supporting that this model can be able to differentiate between the two well. With the total of false positives being 174 and false negatives being 178, it shows a close even ratio, but the amount of both incorrect predictions makes its performance decent, but not the best model for this experiment.

As the overall scores for this method were decent, this may be due to the Word2Vec not being able to identify slang, potential misspellings, and new words as it was trained on Google News articles from 2013 which focuses on clean and professional vocabulary, and may affect the outcomes, as the model may not understand them as well as it could. We had trained the Word2Vec model on the Google News data as it is a widely cited and used for fake news detection or other instances similar to this project. We had also tried using a Word2Vec model training on twitter data specifically, but it performed poorly compared to the Google News Word2Vec model. In addition to this, Word2Vec does not handle short tweets well, which could affect the values and outcome. Through the use of the mean pool, this could cause the tweets to lose more meaning as they are averaged out, lose the order, and be isolated without knowing the true context. As I have used the Random Forest classifier, this may have also affected the results, as it does not utilize semantic relationships between words. This classifier cannot handle multiple embedding dimensions well, which can cause this method to miss some connections between certain information.

6.3 GloVe + Decision Tree

The GloVe + Decision Tree model achieved an accuracy of 0.6590, precision of 0.6746, recall of 0.6602, F1-score of 0.6673, and ROC-AUC of 0.6590 on the hold-out test set. The 5-fold cross-validation results were similar, with an average accuracy of 0.6625, showing that the model performance was stable.

From the confusion matrix, the model correctly classified 634 fake tweets and 684 true tweets, but it also made a noticeable number of errors, including 330 false positives and 352 false negatives. This indicates that the model struggled to clearly separate fake and true tweets compared to stronger approaches.

One reason for this lower performance is that GloVe uses fixed word embeddings, which do not fully capture the context of words in a sentence. Also, averaging word vectors can lose important information like word order and sentence structure. Compared with BERT-based methods, which use contextual embeddings, the GloVe model is less effective at understanding nuanced language patterns. However, it still provides a useful baseline because it incorporates semantic information while remaining relatively simple and efficient.

6.4 BERT (Fine-Tuned) + Classification Head

The BERT fine-tuned model with a classification head achieved strong results for fake news detection. It reached an accuracy of 0.9330, precision of 0.9312, recall of 0.9402, F1-score of 0.9356, and ROC-AUC score of 0.9808. These results show that the model performed very well overall and had an excellent ability to distinguish between fake and true news. However, it was not the best-performing model overall, since TF-IDF + Logistic Regression achieved higher accuracy and F1-score.

From the confusion matrix, the model correctly classified most fake and true news samples, with only a small number of false positives and false negatives. This indicates that the model had a good balance between precision and recall and did not strongly favor one class over the other.

The strong performance can be explained by fine-tuning, which allows BERT to adapt its contextual language understanding to the specific fake news detection task. Unlike using BERT only as a fixed embedding extractor, fine-tuned BERT can learn task-specific patterns and semantic relationships in the text.

6.5 BERT Embeddings + SVM

BERT embeddings + SVM achieved a decent performance, with an accuracy of 0.7725 and an average macro F1-score of 0.7721, showing that it maintained a balanced performance for both fake and true news classes. Within the true class, the model had a precision of 0.7764, a recall of 0.7876, and an F1-score of 0.7820. This shows that it was able to identify a fair number of true news instances. Similarly, the fake news class had a precision of 0.7682, a recall of 0.7562, and an F1-score of 0.7622, showing only a slightly worse performance. The ROC-AUC score of 0.8580 indicates that the model has a good ability to distinguish between fake and true news.

Looking at the confusion matrix, the model correctly classified 729 fake tweets as fake and 816 true tweets as true. 220 true tweets were predicted as fake, and 235 fake tweets were predicted to be true.

Bert likely underperforms here because raw sequence embeddings from a base model don't perfectly highlight the specific linguistics of fake news without fine-tuning. If we look at the BERT fine-tuned, we can see far better results, although it was done with a classification head rather than SVM, so it can't be a direct comparison. But overall, BERT embeddings with SVM had good performance, performing far better than random chance.

6.6 Raw Text LLM Prompting

In contrast to the traditional ML models using a sample of 10,000 tweets, a sample of 1,000 tweets was used in the testing of the LLM due to API token limits. The samples were still left unbalanced from the same dataset. Note that 999 tweets were able to be processed successfully. The remaining 1 test failed due to the LLM providing an infinite looping response.

LLM Prompting did not perform well, obtaining an accuracy of 0.6356, a precision of 0.6360, a recall of 0.6343, and an F1-score of 0.6351. These results suggest that direct prompting alone was less effective at distinguishing fake and true tweets in this dataset. ROC-AUC was not calculated for this method because the model produced deterministic class labels rather than continuous confidence scores. ROC-AUC is most useful for classifiers that output probabilities and can be evaluated across multiple decision thresholds.

One likely reason for the lower performance is that the LLM was used in a zero-shot setting and did not learn from previous examples during classification. Unlike supervised models such as SVM or fine-tuned BERT, the prompted LLM was not explicitly trained on this dataset and has no opportunity to adapt to dataset-specific patterns. The smaller evaluation sample size and token constraints may have also limited the consistency and depth of the model's reasoning.

Looking into specific examples classified incorrectly, one tweet classified as fake that is actually true is "never forget, Matt Santos ran on Medicare for All in 2005, and won the Presidency. That's my President." The LLM's reasoning explained how Matt Santos is a fictional character who ran for president and won in a fictional world. The tweet is false since he didn't win the real presidency of the United States. This highlights the LLM's inability to understand the context of tweets.

An example of the LLM classifying a false tweet as true was a tweet about COVID-19. The reasoning behind the predicted true label was that "The post appears to be a summary of a scientific study or research findings on COVID-19 transmission, which is a credible topic. The language used is formal and objective, suggesting a reliable source." This shows that the LLM believed the tweet to be true due to the language sounding credible, not because it was credible. This reasoning is not optimal for detecting fake news and misinformation.

7 Discussion and Comparison

The experimental results demonstrate that traditional ML models and fine-tuned transformers outperformed LLM-based prompting under this experimental setup. From the six methods, TF-IDF + Logistic Regression was the best implementation with an accuracy of 0.9575 and an ROC-AUC of 0.9873. This suggests that specific keyword patterns and phrasings are a good indication of credibility.

BERT (fine-tuned) followed closely with an accuracy of 0.933, out performing BERT (embeddings) significantly, which had an accuracy of 0.7725. Embeddings likely underperformed fine-tuned BERT because raw sequence embeddings from a base model don't perfectly highlight the linguistics of fake news.

Word2Vec and GloVe's performance reveals the limitations of mean pooling. Word2Vec achieved an accuracy of 0.8240, and GloVe achieved an accuracy of 0.6590. Averaging word vectors caused the model to lose the specific word order and overall sentence structure. Word2Vec also could have struggled due to its older training data, which meant it lacked modern slang. Glove's struggle, on the other hand came from its reliance on fixed word embeddings.

LLM prompting's score was the lowest of all six methods, with an accuracy of 0.6356. The reliance on zero-shot reasoning meant the model could have mislabeled a tweet simply because the language sounded formal and objective. However, it did still perform better than random chance, which gives potential for better performance given either more training and/or a different model.

It is important to note the constraints associated with the LLM prompting method used in this study. Because only 1,000 tweets were evaluated instead of 10,000, the reported metrics may be less stable and less representative of the full dataset. With fewer examples, performance can vary depending on which tweets were randomly selected, while a larger sample generally provides a more reliable estimate of model performance. Another limitation was the API usage constraints on Cerebras.ai. The daily token quota limited the total number of tweets that could be processed within reasonable time and resource limits. As a result, input tweet text was limited to a maximum of 500 characters, and each response was limited to 256 output tokens. These restrictions may have reduced the depth of reasoning of the model and the length of its explanations.

8 Conclusion

In this paper, we have discussed the importance of identifying fake news within social media platforms along with our approach to finding the best method of the six proposed methods to differentiate between fake news and authentic news. With these six methods, we have displayed and evaluated the results in terms of how well each performed and its limitations. Overall, we have concluded that the TF-IDF + Logistic Regression method performed best among the six proposed methods, as shown by the results and confusion matrices.

9 References

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Table 2: Performance Comparison of Six Methods

Method	Accuracy	Precision	Recall	F1-score	ROC-AUC
TF-IDF + Logistic Regression	0.9575	0.9579	0.9571	0.9574	0.9873
Word2Vec + Random Forest	0.8240	0.8314	0.8282	0.8298	0.8997
GloVe + Decision Tree	0.6590	0.6746	0.6602	0.6673	0.6590
BERT (fine-tuned) + Classification Head	0.933	0.931	0.940	0.936	0.9808
BERT (embeddings) + SVM	0.7725	0.7764	0.7876	0.7820	0.8580
LLM Prompting	0.6356	0.6360	0.6343	0.6351	N/A

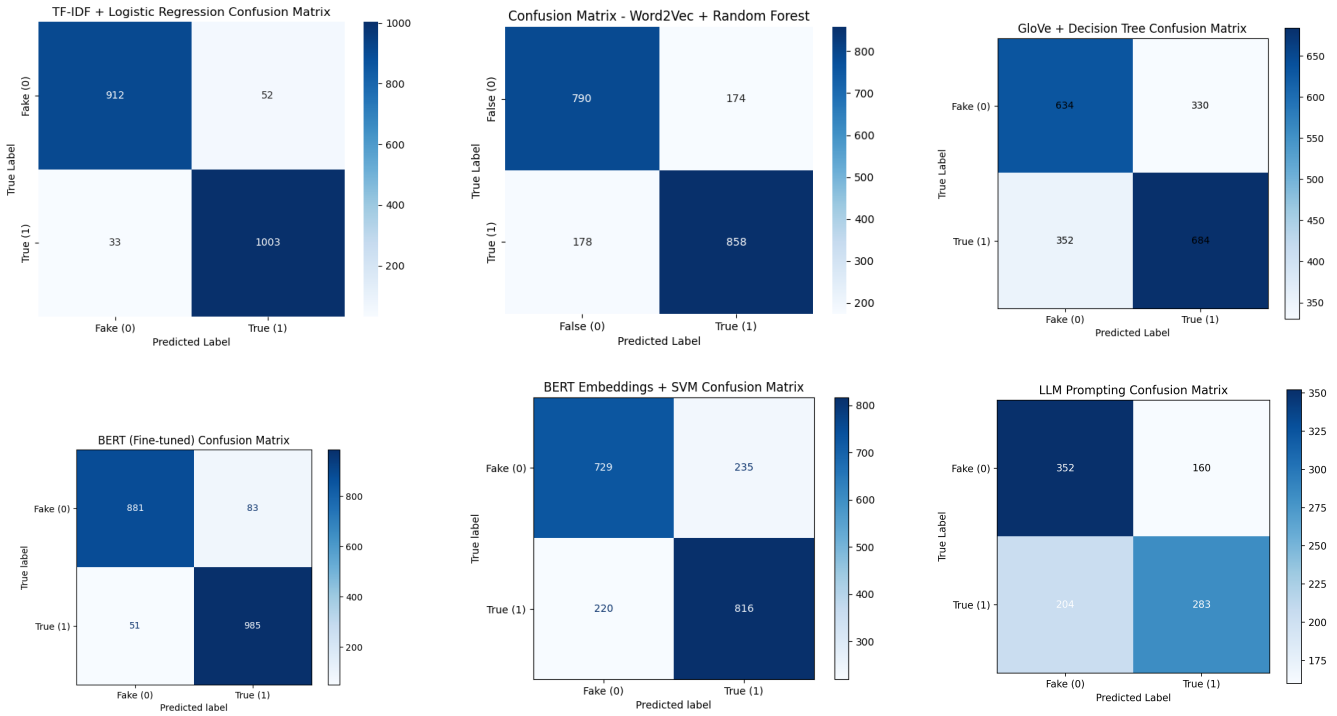

Figure 1: Confusion matrices for all six fake news detection methods.